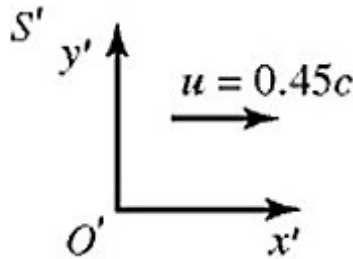
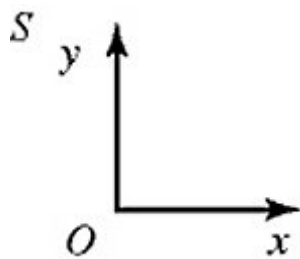


MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

- 1) A star is moving towards the earth with a speed at 90% the speed of light. It emits light, which moves away from the star at the speed of light. Relative to us on earth, what is the speed of the light moving toward us from the star? 1) _____
 A) $1.9c$ B) $0.90c$ C) c D) $1.1c$ E) $1.20c$
- 2) As measured in Earth's rest frame, a spaceship traveling at $0.964c$ takes 12.2 y to travel between planets. How long does the trip take as measured by someone on the spaceship? 2) _____
 A) 7.93 y B) 33.4 y C) 3.24 y D) 45.9 y
- 3) An astronaut on a spaceship moving at $0.927c$ says that the trip between two stationary stars took 8.04 y. How long does this journey take as measured by someone at rest relative to the two stars? 3) _____
 A) 21.4 y B) 4.35 y C) 3.02 y D) 23.7 y
- 4) Relative to the frame of the observer making the measurement, at what speed parallel to its length is the length of a meterstick 60 cm? 4) _____
 A) $0.70c$ B) $0.80c$ C) $0.90c$ D) $0.60c$ E) $0.50c$
- 5) A particle in a 748 m-long linear particle accelerator is moving at $0.875c$. How long does the particle accelerator appear to the particle? 5) _____
 A) 972 m B) 1545 m C) 362 m D) 172 m
- 6) System S' has a velocity $u = +0.45c$ relative to system S , as shown in the figure. The clocks of S and S' are synchronized at $t = t' = 0$ when the origins O and O' coincide. An event is observed in both systems. The event takes place at $x = 600$ m and at time $t = 1.9 \mu\text{s}$, as measured by an observer in S . What is the x' -coordinate of the event, measured by an observer in S' ? 6) _____



- A) 380 m B) 310 m C) 360 m D) 350 m E) 340 m

- 7) A navigational beacon in deep space broadcasts at a radio frequency of 50 MHz. A spaceship approaches the beacon with a relative velocity of $0.40c$. What is the frequency of the beacon radio signal that is detected on the ship? 7) _____
 A) 60 MHz B) 55 MHz C) 66 MHz D) 76 MHz E) 71 MHz
- 8) In a certain particle accelerator, a proton has a kinetic energy that is equal to its rest energy. What is the speed of the proton relative to the accelerator? 8) _____
 A) $0.87c$ B) $0.71c$ C) $0.75c$ D) $0.25c$ E) $0.50c$
- 9) In the lab, a relativistic proton has a momentum of $1.00 \times 10^{-19} \text{ kg} \cdot \text{m/s}$ and a rest energy of 0.150 nJ. What is the speed of the proton in the lab? ($c = 3.00 \times 10^8 \text{ m/s}$, $m_{\text{proton}} = 1.67 \times 10^{-27} \text{ kg}$) 9) _____
 A) $0.196c$ B) $0.911c$ C) $0.951c$ D) $0.170c$ E) $0.930c$
- 10) How many joules of energy are required to accelerate a 1.0-kg mass from rest to a speed of 86.6% the speed of light? ($c = 3.00 \times 10^8 \text{ m/s}$) 10) _____
 A) $1.8 \times 10^{17} \text{ J}$
 B) $4.5 \times 10^9 \text{ J}$
 C) $3.0 \times 10^3 \text{ J}$
 D) $9.0 \times 10^{16} \text{ J}$
 E) $2.7 \times 10^{12} \text{ J}$
- 11) During a nuclear reaction, $1.7 \times 10^{-4} \text{ J}$ of energy is released. What is the resulting change in mass of the particles involved? ($c = 3.00 \times 10^8 \text{ m/s}$) 11) _____
 A) $1.5 \times 10^{-13} \text{ kg}$
 B) $4.8 \times 10^{-18} \text{ kg}$
 C) $5.1 \times 10^{-4} \text{ kg}$
 D) $1.9 \times 10^{-21} \text{ kg}$
 E) $4.3 \times 10^{-11} \text{ kg}$
- 12) A relativistic proton has a momentum of $1.0 \times 10^{-17} \text{ kg} \cdot \text{m/s}$ and a rest energy of 0.15 nJ. What is the kinetic energy of this proton? ($c = 3.00 \times 10^8 \text{ m/s}$, $m_{\text{proton}} = 1.67 \times 10^{-27} \text{ kg}$) 12) _____
 A) 1.3 pJ B) 2.5 pJ C) 3.0 pJ D) 2.2 pJ E) 1.6 pJ

- 13) If the accuracy in measuring the velocity of a particle increases, the accuracy in measuring its position will _____
A) increase.
B) remain the same.
C) decrease.
D) It is impossible to say since the two measurements are independent and do not affect each other.
- 14) Light of wavelength 400 nm falls on a metal surface having a work function 1.70 eV. What is the maximum kinetic energy of the photoelectrons emitted from the metal? ($c = 3.00 \times 10^8$ m/s, $h = 6.626 \times 10^{-34}$ J $\cdot$ s = 4.141×10^{-15} eV $\cdot$ s, 1 eV = 1.60×10^{-19} J)
A) 2.82 eV B) 4.52 eV C) 1.41 eV D) 1.70 eV E) 3.11 eV
- 15) When a certain metal is illuminated by light, photoelectrons are observed provided that the wavelength of the light is less than 473 nm. Which one of the following values is closest to the work function of this metal? ($h = 6.626 \times 10^{-34}$ J $\cdot$ s, $c = 3.00 \times 10^8$ m/s, 1 eV = 1.60×10^{-19} J)
A) 2.8 eV B) 2.6 eV C) 3.1 eV D) 3.3 eV
- 16) A metal having a work function of 2.4 eV is illuminated with monochromatic light whose photon energy is 4.1 eV. What is the maximum kinetic energy of the photoelectrons produced by this light? ($h = 6.626 \times 10^{-34}$ J $\cdot$ s, 1 eV = 1.60×10^{-19} J)
A) 2.7×10^{-19} J
B) 4.7×10^{-19} J
C) 3.8×10^{-19} J
D) 6.6×10^{-19} J
E) 5.6×10^{-19} J
- 17) In a photoelectric effect experiment, electrons emerge from a copper surface with a maximum kinetic energy of 2.70 eV when light shines on the surface. The work function of copper is 4.65 eV. Which one of the following values is closest to the wavelength of the light? ($h = 6.626 \times 10^{-34}$ J $\cdot$ s, $c = 3.00 \times 10^8$ m/s, 1 eV = 1.60×10^{-19} J)
A) 460 nm B) 250 nm C) 130 nm D) 170 nm

- 18) A laser emits light of wavelength 359 nm during a brief pulse that lasts for 25 ms and has a total energy of 1.0 J. How many photons are emitted in that single pulse? 18) _____
 ($c = 3.00 \times 10^8$ m/s, $h = 6.626 \times 10^{-34}$ J • s)
 A) 2.2×10^{19}
 B) 1.4×10^{17}
 C) 7.2×10^{16}
 D) 1.8×10^{18}
 E) 4.5×10^{19}
- 19) A light beam from a 3.7-mW He-Ne laser has a wavelength of 633 nm. How many photons does the laser emit in one second? ($h = 6.626 \times 10^{-34}$ J • s, $c = 3.00 \times 10^8$ m/s) 19) _____
 A) 1.9×10^{16} B) 2.3×10^{16} C) 1.2×10^{16} D) 1.5×10^{16}
- 20) A photon of wavelength 18.0 pm is scattered through an angle of 120° by a stationary electron. What is the wavelength of the scattered photon? ($m_e = 9.11 \times 10^{-31}$ kg, $h = 6.626 \times 10^{-34}$ J • s, $c = 3.00 \times 10^8$ m/s) 20) _____
 A) 22.9 pm B) 19.2 pm C) 20.4 pm D) 24.1 pm E) 21.6 pm
- 21) A nonrelativistic electron and a nonrelativistic proton have the same de Broglie wavelength. Which of the following statements about these particles are accurate? 21) _____
 (There may be more than one correct choice.)
 A) The electron has more momentum than the proton.
 B) The electron has more kinetic energy than the proton.
 C) Both particles have the same momentum.
 D) Both particles have the same kinetic energy.
 E) Both particles have the same speed.
- 22) At absolute temperature T , a black body radiates its peak intensity at wavelength λ . At absolute temperature $2T$, what would be the wavelength of the peak intensity? 22) _____
 A) $\lambda/2$ B) λ C) $\lambda/16$ D) 2λ E) 16λ
- 23) Monochromatic light strikes a metal surface and electrons are ejected from the metal. If the intensity of the light is increased, what will happen to the ejection rate and maximum energy of the electrons? 23) _____
 A) greater ejection rate; greater maximum energy
 B) same ejection rate; greater maximum energy
 C) same ejection rate; same maximum energy
 D) greater ejection rate; same maximum energy

- 24) The energy of the ground state in the Bohr model of the hydrogen atom is -13.6 eV. In a transition from the $n = 2$ state to the $n = 4$ state, a photon of energy 24) _____
- A) 0.85 eV is absorbed.
 B) 2.55 eV is emitted.
 C) 3.40 eV is emitted.
 D) 3.40 eV is absorbed.
 E) 2.55 eV is absorbed.
- 25) A hydrogen atom initially in the $n = 4$ state decays to the $n = 2$ state. The emitted photon is detected in a photographic plate. What is the wavelength of the detected photon? The lowest level energy state of hydrogen is -13.6 eV. ($h = 6.626 \times 10^{-34} \text{ J} \cdot \text{s}$, $1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$, $c = 3.00 \times 10^8 \text{ m/s}$) 25) _____
- A) 95.0 nm B) 486 nm C) 1880 nm D) 97.3 nm
- 26) What is the energy of a photon that has a wavelength equal to the de Broglie wavelength of a proton having a speed of $7.0 \times 10^4 \text{ m/s}$? ($m_{\text{proton}} = 1.67 \times 10^{-27} \text{ kg}$, $c = 3.00 \times 10^8 \text{ m/s}$) 26) _____
- A) 360 keV B) 430 keV C) 220 keV D) 290 keV E) 150 keV
- 27) What is the frequency of the light emitted by atomic hydrogen with $m = 8$ and $n = 12$? (The Rydberg constant is $R = 1.097 \times 10^7 \text{ m}^{-1}$, $c = 3.00 \times 10^8 \text{ m/s}$.) 27) _____
- A) $7.46 \times 10^{13} \text{ Hz}$
 B) $2.86 \times 10^{13} \text{ Hz}$
 C) $1.05 \times 10^{13} \text{ Hz}$
 D) $8.82 \times 10^{13} \text{ Hz}$
 E) $1.43 \times 10^{13} \text{ Hz}$
- 28) What is the orbital radius of the $n = 3$ excited state in the Bohr model of the hydrogen atom? The ground-state radius of the hydrogen atom is $0.529 \times 10^{-10} \text{ m}$. 28) _____
- A) 0.477 nm B) 0.549 nm C) 0.159 nm D) 0.382 nm
- 29) What is the wavelength of peak emission for a black body at 37°C ? ($c = 3.0 \times 10^8 \text{ m/s}$, Wien displacement law constant is $2.9 \times 10^{-3} \text{ m} \cdot \text{K}$, $\sigma = 5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4$) 29) _____
- A) 94 μm B) 29 μm C) 9.4 μm D) 7.8 μm E) 78 μm

30) A nonrelativistic electron is confined to a length of 500 pm on the x -axis. What is the kinetic energy of the electron if its speed is equal to the minimum uncertainty possible in its speed? ($\hbar = 1.055 \times 10^{-34} \text{ J} \cdot \text{s}$, $m_{\text{el}} = 9.11 \times 10^{-31} \text{ kg}$, $1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$)

30) _____

- A) 0.38 eV
- B) 3.8 eV
- C) 0.038 eV
- D) 0.0038 eV
- E) 0.00038 eV

Answer Key

Testname: PHYS224 MODERN PHYSICS MIDTERM

- 1) C
- 2) C
- 3) A
- 4) B
- 5) C
- 6) A
- 7) D
- 8) A
- 9) A
- 10) D
- 11) D
- 12) C
- 13) C
- 14) C
- 15) B
- 16) A
- 17) D
- 18) D
- 19) C
- 20) E
- 21) B, C
- 22) A
- 23) D
- 24) E
- 25) B
- 26) C
- 27) B
- 28) A
- 29) C
- 30) C